ERC-4626 Yield Vault with Multi-Criteria Decision Making

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1. Overview

This project implements an ERC-4626 tokenized vault that automatically rebalances user deposits between Aave V3 and Compound V3 lending protocols. An off-chain agent evaluates protocols using a weighted multi-criteria scoring model and submits cryptographically signed rebalance decisions to the on-chain vault. A Chainlink Automation fallback ensures the vault remains managed if the agent goes offline.

2. Architecture

The system has two layers:

- **Off-chain agent (Python):** Reads on-chain data every hour, applies EMA smoothing to rates, scores each protocol using 4-factor MCDM, and signs rebalance decisions with EIP-712.
- **On-chain vault (Solidity):** Receives signed parameters, verifies the keeper's ECDSA signature, checks nonce/timestamp/cooldown, and executes the rebalance through protocol adapters.

Contracts: AIVault.sol (ERC-4626 + UUPS proxy), StrategyManager.sol (validation + rate tracking), AaveV3Adapter.sol, CompoundV3Adapter.sol, RateMath.sol (normalization library).

The adapter pattern (`IProtocolAdapter` interface) allows adding new protocols without changing existing contracts.

3. Formulas

Share pricing (ERC-4626 with virtual shares for inflation attack protection):

$$s = \left\lfloor \frac{a \cdot (S + 10^6)}{A + 1} \right\rfloor$$

where a = deposited assets, S = total supply, A = total assets. The 10^6 offset prevents the known ERC-4626 inflation/donation attack.

APY normalization (cross-protocol, to annual 1e18 scale):

- Aave V3: $\text{APY}_{1e18} = \text{liquidityRate}_{RAY}/10^9$ (RAY = 10^{27})
- Compound V3: $\text{APY}_{1e18} = r_{sec} \times 31{,}557{,}600$

EMA smoothing (rate noise and manipulation resistance):

$$S_t = \alpha \cdot R_t + (1 - \alpha) \cdot S_{t-1}, \quad \alpha = 0.3$$

Rate jump guard: if $|R_t - S_{t-1}| > 5\%$, the update is skipped.

MCDM scoring model:

$$\text{Score}_i = 0.40 \cdot f_{APY} + 0.25 \cdot f_{Risk} + 0.20 \cdot f_{Cost} + 0.15 \cdot f_{Stability}$$

Factor	Weight	Meaning
APY	40%	Normalized smoothed yield
Risk	25%	$1 -$ utilization (high utilization = risk)
Cost	20%	$1 -$ gasCost/0.01 (gas efficiency)
Stability	15%	$1 - \|\Delta TVL\|/0.30$ (TVL change penalty)

Decision rule: rebalance when $\text{Score}_{best} - \text{Score}_{current} \geq 0.05$.

On-chain fallback threshold (Chainlink path):

$$\text{rebalance if } \frac{\Delta APY \times TVL \times t_{since}}{365 \text{ days}} > \text{gasCost}$$

4. Security

Threat	Mitigation
Inflation attack	Virtual shares (10^6 offset in `_decimalsOffset`)
Reentrancy	OpenZeppelin ReentrancyGuard on all external calls
Rate manipulation	Dual EMA (on-chain + agent) + 5% jump guard
Signature forgery	EIP-712 typed data + ECDSA verification
Replay attack	Sequential nonce + 5-minute timestamp TTL
Agent downtime	Chainlink Automation fallback (6-hour timeout)
Rapid exploitation	1-hour cooldown between rebalances

Access control: deposits/withdrawals are open; rebalance requires a valid keeper signature; emergency functions are owner-only; pause/unpause is owner-only.

5. Testing

67 tests total, all passing:

- **37 Solidity unit tests** (concrete + fuzz with 1000 runs each)
- **4 integration tests** (full lifecycle: deposit -> rebalance -> yield -> withdraw)
- **6 invariant tests** (stateful fuzzing: 76,800+ random call sequences, 0 violations)
- **20 Python scoring tests** (boundary conditions, weight interactions)

Key invariants verified: vault solvency, deposit/withdrawal accounting, share conversion consistency, non-decreasing share price.

6. Deployment

Deployed on Ethereum Sepolia (chain 11155111). All contracts verified on Sourcify. UUPS proxy pattern (ERC-1967) enables future upgrades without fund migration.

References

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4. Compound III – https://docs.compound.finance/
5. Chainlink Automation – https://docs.chain.link/chainlink-automation
6. OpenZeppelin Contracts v5 – https://docs.openzeppelin.com/contracts/5.x/